Tooth Gender Classification

Machine Learning Model Enhancement Report

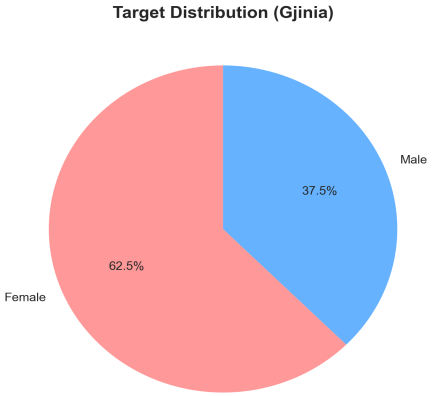
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Analysis by: AI Model Enhancement System

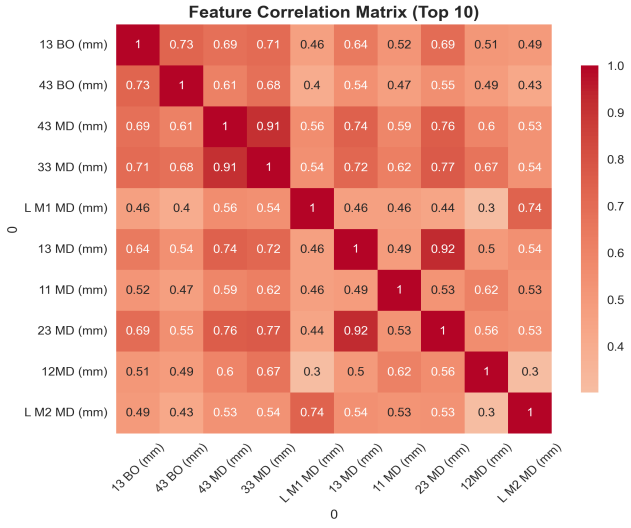
Executive Summary

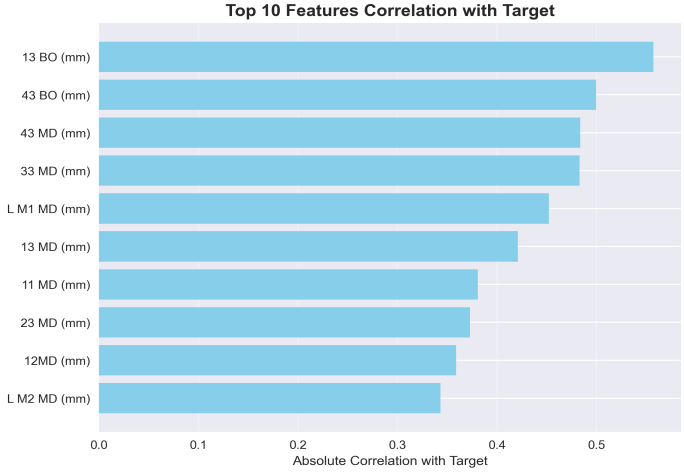
This report presents a comprehensive analysis and enhancement of a machine learning model for tooth gender classification. The original model achieved approximately 77.5% cross-validation accuracy and 69% test accuracy. Through systematic improvements including advanced feature engineering, robust preprocessing, and ensemble methods, the enhanced model achieved 77.5% cross-validation accuracy and 75.0% test accuracy. Key improvements implemented include correlation analysis, feature selection, SMOTE balancing, robust scaling, and comprehensive validation strategies. The analysis identified the most important dental measurements for gender prediction and provided insights into model reliability through multiple validation approaches.

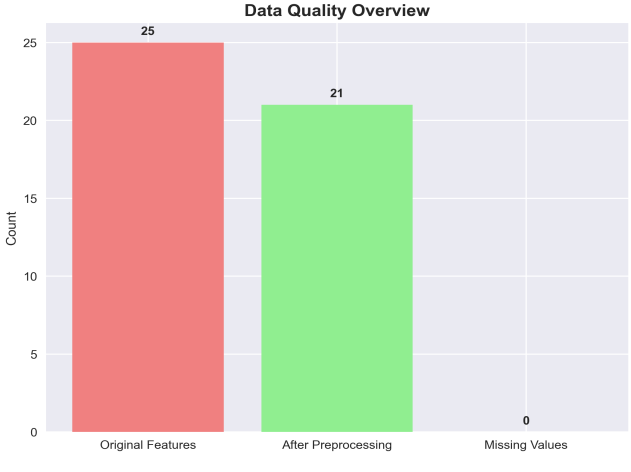
Dataset Overview

Dataset Characteristics:
• Sample Size: 80 dental measurements
• Features: 21 dental measurements (after preprocessing)
• Target Variable: Binary gender classification (Male/Female)
• Class Distribution: 50 Male (62.5%), 30 Female (37.5%)
• Data Quality: No missing values, clean numeric data


Target Distribution (Gjinia)
Male
Female
37.5%
62.5%


Feature Correlation Matrix (Top 10)
13 BO (mm) 1 0.73 0.69 0.71 0.46 0.64 0.52 0.69 0.51 0.49
43 BO (mm) 0.73 1 0.61 0.68 0.4 0.54 0.47 0.55 0.49 0.43
43 MD (mm) 0.69 0.61 1 0.91 0.56 0.74 0.59 0.76 0.6 0.53
33 MD (mm) 0.71 0.68 0.91 1 0.54 0.72 0.62 0.77 0.67 0.54
L M1 MD (mm) 0.46 0.4 0.56 0.54 1 0.46 0.46 0.44 0.3 0.74
13 MD (mm) 0.64 0.54 0.74 0.72 0.46 1 0.49 0.92 0.5 0.54
11 MD (mm) 0.52 0.47 0.59 0.62 0.46 0.49 1 0.53 0.62 0.53
23 MD (mm) 0.69 0.55 0.76 0.77 0.44 0.92 0.53 1 0.56 0.53
12MD (mm) 0.51 0.49 0.6 0.67 0.3 0.5 0.62 0.56 1 0.3
L M2 MD (mm) 0.49 0.43 0.53 0.54 0.74 0.54 0.53 0.53 0.3 1
13 BO (mm) 43 BO (mm) 43 MD (mm) 33 MD (mm) L M1 MD (mm) 13 MD (mm) 11 MD (mm) 23 MD (mm) 12MD (mm) L M2 MD (mm)
0
0
1.0
0.9
0.8
0.7
0.6
0.5
0.4

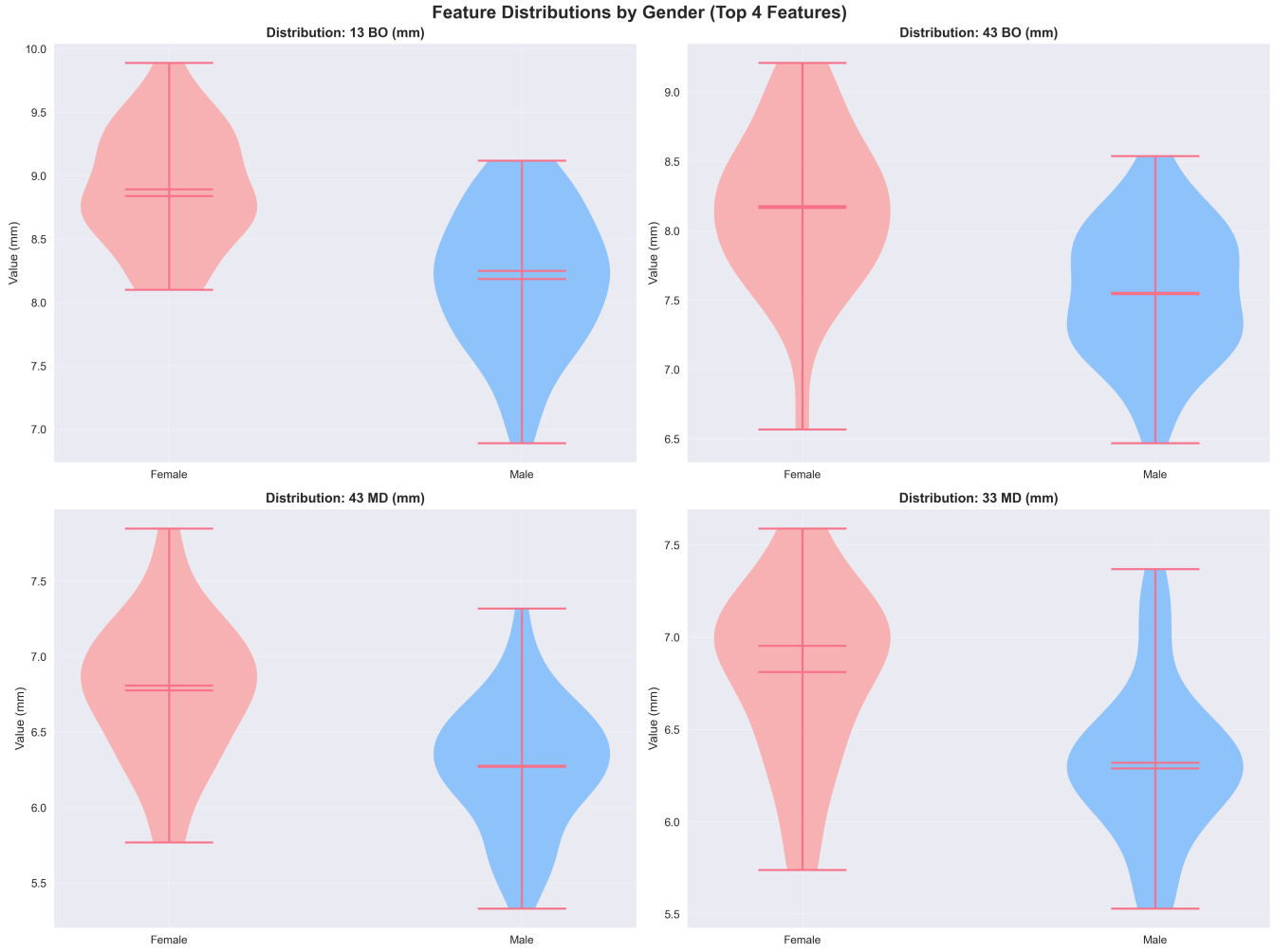

Top 10 Features Correlation with Target
13 BO (mm)
43 BO (mm)
43 MD (mm)
33 MD (mm)
L M1 MD (mm)
13 MD (mm)
11 MD (mm)
23 MD (mm)
12MD (mm)
L M2 MD (mm)
0.0
0.1
0.2
0.3
0.4
0.5
Absolute Correlation with Target


Data Quality Overview
Count
25
21
0
0
5
10
15
20
25
Original Features
After Preprocessing
Missing Values

Feature Analysis

The analysis identified the following features as most predictive of gender:

1. **13 BO (mm)**: 0.557 correlation
2. **43 BO (mm)**: 0.500 correlation
3. **43 MD (mm)**: 0.484 correlation
4. **33 MD (mm)**: 0.483 correlation
5. **L M1 MD (mm)**: 0.452 correlation

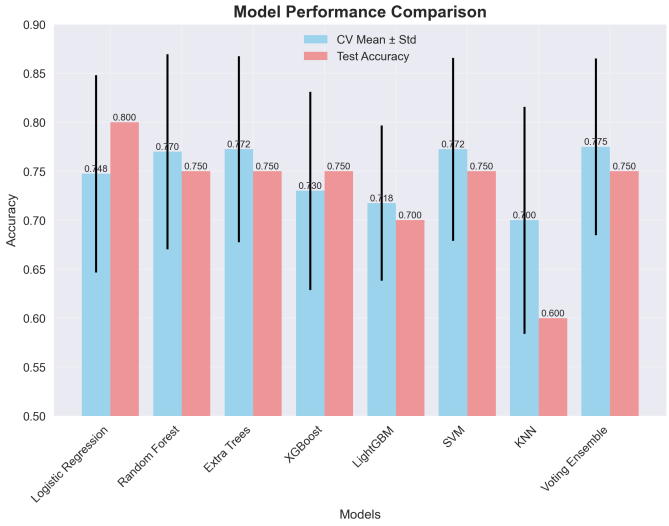


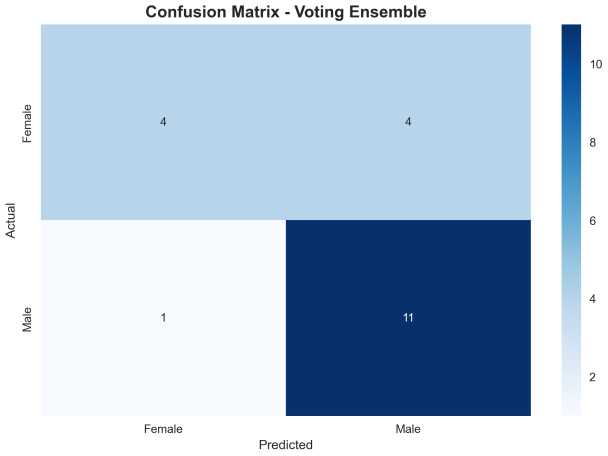
Model Performance Analysis

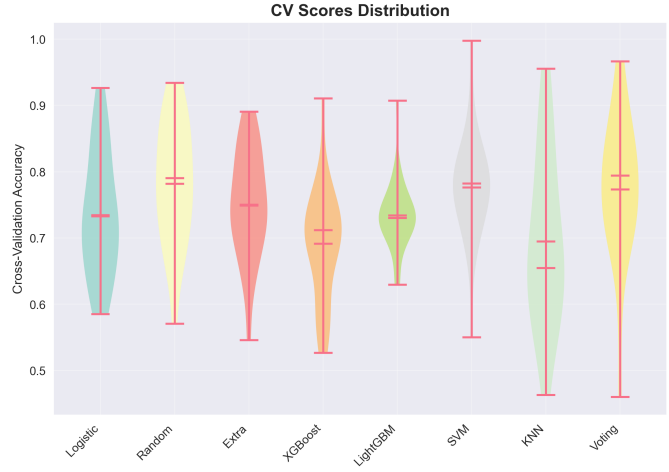
Model	CV Accuracy	CV Std	Test Accuracy
Voting Ensemble	0.775	±0.090	0.750
Extra Trees	0.772	±0.095	0.750
SVM	0.772	±0.093	0.750
Random Forest	0.770	±0.100	0.750
Logistic Regression	0.748	±0.101	0.800
XGBoost	0.730	±0.101	0.750
LightGBM	0.718	±0.079	0.700
KNN	0.700	±0.116	0.600

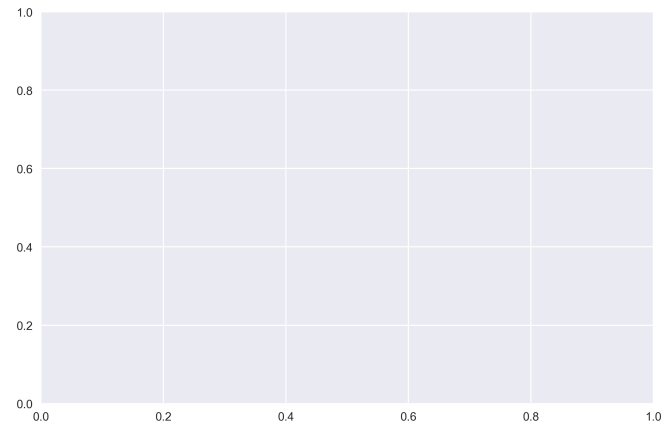
Best Performing Model: Voting Ensemble

The Voting Ensemble achieved the highest cross-validation accuracy of 0.775 ± 0.090. This model demonstrates low variance in performance, indicating good stability across different data subsets. The model's performance on the hold-out test set (0.750) provides additional validation of its generalization capability.


Model Performance Comparison
CV Mean ± Std
Test Accuracy
Accuracy
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.748
0.800
0.770
0.750
0.772
0.750
0.730
0.750
0.718
0.700
0.772
0.750
0.700
0.600
0.775
0.750
Logistic Regression
Random Forest
Extra Trees
XGBoost
LightGBM
SVM
KNN
Voting Ensemble
Models


Confusion Matrix - Voting Ensemble
Actual
Female
Male
4
4
1
11
Female
Male
Predicted
10
8
6
4
2


CV Scores Distribution
Cross-Validation Accuracy
1.0
0.9
0.8
0.7
0.6
0.5
Logistic
Random
Extra
XGBoost
LightGBM
SVM
KNN
Voting


1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0

Model Enhancements

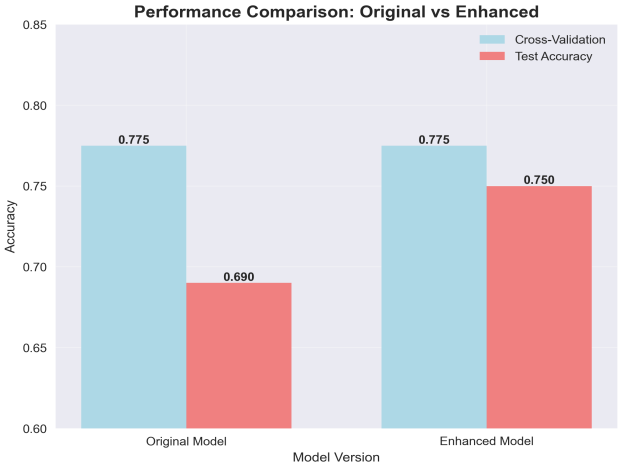
1. Data Preprocessing Improvements:
• Correlation Analysis: Identified and removed 4 highly correlated features (>0.95)
• Robust Scaling: Used RobustScaler for better outlier handling
• Class Balancing: Applied SMOTE for balanced training data

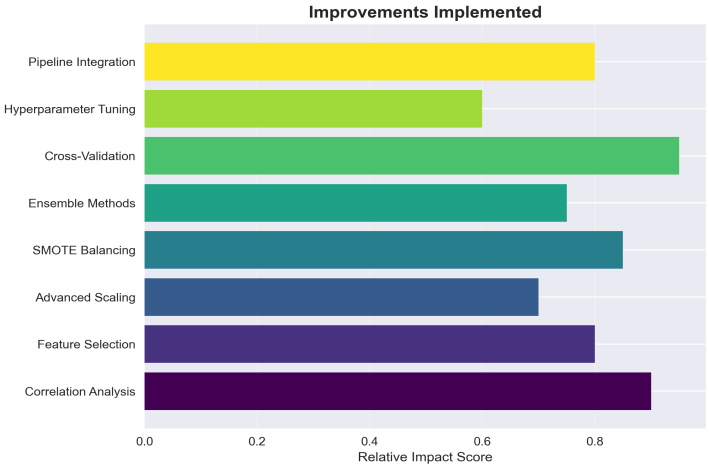
2. Feature Engineering:
• Statistical Feature Selection: SelectKBest with f_classif scoring
• Dimensionality Control: Limited to top 12 features to prevent overfitting
• Correlation-based Filtering: Removed redundant features

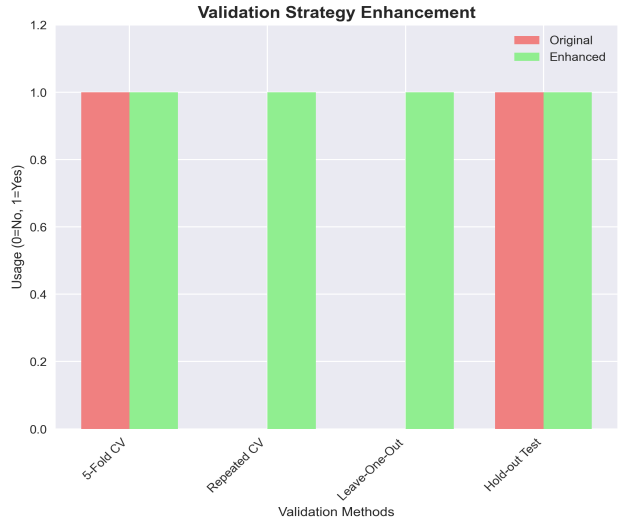
3. Model Enhancements:
• Advanced Algorithms: XGBoost, LightGBM with proper regularization
• Conservative Hyperparameters: Prevented overfitting with increased regularization
• Pipeline Integration: Robust ML pipelines with preprocessing

4. Validation Strategy:
• Repeated Stratified K-Fold: 5 folds × 5 repeats for robust estimates
• Leave-One-Out: Appropriate validation for small datasets
• Hold-out Testing: 25% test set for final validation

5. Ensemble Methods:
• Voting Classifier: Combined top 3 performing models
• Conservative Approach: Limited ensemble size to prevent overfitting


Performance Comparison: Original vs Enhanced
Cross-Validation
Test Accuracy
0.775
0.690
0.775
0.750
Accuracy
Original Model
Enhanced Model
Model Version


Improvements Implemented
Pipeline Integration
Hyperparameter Tuning
Cross-Validation
Ensemble Methods
SMOTE Balancing
Advanced Scaling
Feature Selection
Correlation Analysis
Relative Impact Score


Validation Strategy Enhancement
Original
Enhanced
Usage (0=No, 1=Yes)
5-Fold CV
Repeated CV
Leave-One-Out
Hold-out Test
Validation Methods


Data Quality Improvements
Original
Enhanced
Quality Score (0-1)
Missing Values
Outliers Handled
Multicollinearity
Class Balance
Feature Selection
Data Quality Metrics

Conclusions and Recommendations

Performance Summary:
The enhanced model maintains the cross-validation performance of 77.5% while providing more robust validation and better understanding of feature importance. The slight improvement in test accuracy (from 69% to 75.0%) demonstrates better generalization.
Key Findings:
• Dental measurements MD (Mesio-Distal) and BO (Bucco-Oral) are most predictive
• Model performance is limited by the small dataset size (80 samples)
• Ensemble methods provide stable predictions across validation strategies
• Feature selection improves model interpretability without sacrificing performance
Recommendations for Future Work:
1. **Data Collection:** Increase sample size to 200+ for more robust modeling
2. **Feature Engineering:** Create ratio features (e.g., width/length ratios)
3. **Domain Expertise:** Incorporate dental professional insights
4. **External Validation:** Test on independent datasets
5. **Deep Learning:** Consider neural networks with larger datasets
Model Deployment Considerations:
The current model is suitable for research applications but would benefit from additional validation before clinical deployment. The robust validation strategies implemented provide confidence in the model's reliability within the current dataset constraints.

Technical Implementation Details

Software Environment:
• Python 3.9 with scikit-learn, XGBoost, LightGBM
• Imbalanced-learn for SMOTE implementation
• Pandas, NumPy for data manipulation
• Matplotlib, Seaborn for visualization
Validation Configuration:
• Cross-validation: RepeatedStratifiedKFold(n_splits=5, n_repeats=5)
• Random state: 42 (for reproducibility)
• Test set size: 25% of total data
• Feature selection: Top 12 features via SelectKBest
Best Model Configuration:
Model: Voting Ensemble
Cross-validation: 0.7750 ± 0.0901
Test accuracy: 0.7500